

MATH 8090:

ARMA Models: Prediction and Forecasting

Readings: CC08 Chapter 9; BD16 Chapters 2.5 and 3.3; SS17
Chapter 3.4

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Week 7

Agenda

Linear Predictor

Prediction Equations

ARMA Case Study

Learning Objectives

By the end of this lecture, you should be able to:

- ▶ Explain the distinction between an optimal predictor and a best linear predictor
- ▶ Derive the prediction equations using the orthogonality principle
- ▶ Compute the minimum mean squared prediction error
- ▶ Construct forecasts and prediction intervals for ARMA processes
- ▶ Apply ARMA forecasting to an observed time series

Forecasting Stationary Time Series

Let $\{X_t\}$ be a **stationary process** with mean μ and ACVF $\gamma(\cdot)$.
Based on the observed data, $\mathbf{X}_n = (X_1, X_2, \dots, X_n)^T$, we want to forecast X_{n+h} for some h , a positive integer

- **Question:** What is the best way to do so?
⇒ Need to decide on what “best” means

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- ▶ **Question:** What is the best way to do so?
 $\Rightarrow$ Need to decide on what “best” means
- ▶ A commonly used metric for describing forecast performance is the **mean square prediction error** (MSPE):

$$\text{MSPE} = \mathbb{E} \left[(X_{n+h} - m_n(\mathbf{X}_n))^2 \right].$$

$\Rightarrow$ the best predictor (in terms of MSPE) is

$$m_n(\mathbf{X}_n) = \mathbb{E} [X_{n+h} | \mathbf{X}_n],$$

the **conditional expectation of X_{n+h} given $\mathbf{X}_n$**

Linear Predictor

Calculating $\mathbb{E}[X_{n+h}|\mathbf{X}_n]$ can be difficult in general

- ▶ We will restrict to a linear combination of $X_1, X_2, \dots, X_n$ and a constant $\Rightarrow$ **linear predictor**:

$$\begin{aligned}P_n X_{n+h} &= c_0 + c_1 X_n + c_2 X_{n-1} + \dots + c_n X_1 \\&= c_0 + \sum_{j=1}^n c_j X_{n+1-j}\end{aligned}$$

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- ▶ We select the coefficients that minimize the **h -step-ahead mean squared prediction error**:

$$\mathbb{E} \left([X_{n+h} - P_n X_{n+h}]^2 \right) = \mathbb{E} \left(X_{n+h} - c_0 - \sum_{j=1}^n c_j X_{n+1-j} \right)^2$$

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The best linear predictor is the best predictor if $\{X_t\}$ is Gaussian

How to Determine these Coefficients $\{c_j\}$?

The following steps for calculating the c_j values are the same as those used to calculate **ordinary least squares estimates**

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1. Take the derivative of the MSPE with respect to each coefficient c_j
2. Set each derivative equal to zero
3. Solve with respect to the coefficients

Forecasting Stationary Processes I

For simplicity, let's assume $\mu = 0$ (we can always achieve that by subtracting off μ) so that we don't need the constant term. We have

$$P_n X_{n+h} = c_1 X_n + c_2 X_{n-1} + \cdots + c_n X_1.$$

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We want the MSPE

$$\mathbb{E} \left[(X_{n+h} - P_n X_{n+h})^2 \right] = \mathbb{E} \left[(X_{n+h} - c_1 X_n - c_2 X_{n-1} - \cdots - c_n X_1)^2 \right]$$

as small as possible.

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From now on, let's define

$$\mathbb{E} \left[(X_{n+h} - c_1 X_n - c_2 X_{n-1} - \cdots - c_n X_1)^2 \right] = S(c_1, \cdots, c_n)$$

We are going to take derivative of the $S(c_1, \cdots, c_n)$ with respect to each coefficient c_j

Forecasting Stationary Processes II

S is a quadratic function of $c_1, c_2, \dots, c_n$, so any minimizing set of c_j 's must satisfy these n equations:

$$\frac{\partial S(c_1, \dots, c_n)}{\partial c_j} = 0, \quad j = 1, \dots, n.$$

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Since

$S(c_1, \dots, c_n) = \mathbb{E} [(X_{n+h} - c_1 X_n - c_2 X_{n-1} - \dots - c_n X_1)^2]$, we

have

$$\frac{\partial S(c_1, \dots, c_n)}{\partial c_j} = -2\mathbb{E} \left[\left(X_{n+h} - \sum_{i=1}^n c_i X_{n-i+1} \right) X_{n-j+1} \right] = 0$$

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$$\Rightarrow \text{Cov}(X_{n+h} - \sum_{i=1}^n c_i X_{n-i+1}, X_{n-j+1}) = 0, \quad j = 1, \dots, n$$

$\Rightarrow$ Prediction error is uncorrelated with all RVs used in corresponding predictor

Forecasting Stationary Processes III

Orthogonality principle:

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We have

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We have

$$\text{Cov}(X_{n+h}, X_{n-j+1}) - \sum_{i=1}^n c_i \text{Cov}(X_{n-i+1}, X_{n-j+1}) = 0$$

We obtain $\{c_i; i = 1, \dots, n\}$ by solving the system of linear equations:

$$\left\{ \gamma(h+j-1) = \sum_{i=1}^n c_i \gamma(i-j) : j = 1, \dots, n \right\},$$

to find n unknown c_i 's

Computing $P_n X_{n+h}$ via Matrix Operations

We can rewrite the system of prediction equations as

$$\gamma_n = \Sigma_n \mathbf{c}_n,$$

with $\gamma_n = (\gamma(h), \gamma(h+1), \dots, \gamma(h+n-1))^T$,
 $\mathbf{c}_n = (c_1, c_2, \dots, c_n)^T$ and

$$\Sigma_n = \begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{bmatrix}$$

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Solving for $\mathbf{c}_n$ we have

$$\mathbf{c}_n = \Sigma_n^{-1} \gamma_n$$

Properties of the Prediction Errors

The prediction errors are

$$\begin{aligned}U_{n+h} &= X_{n+h} - P_n X_{n+h} \\&= (X_{n+h} - \mu) - \sum_{j=1}^n c_j (X_{n+1-j} - \mu).\end{aligned}$$

It then follows that

- The prediction error has mean zero

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$$\text{Cov}(U_{n+h}, X_j) = \text{Cov}(X_{n+h} - P_n X_{n+h}, X_j) = 0, \quad j = 1, \dots, n$$

The Minimum Mean Squared Prediction Error

We obtain the minimum value of the MSPE by substituting the expression for c_n into $\mathbb{E} [(X_{n+h} - P_n X_{n+h})^2]$:

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The Minimum Mean Squared Prediction Error (Cont'd)

From the previous slide we have

$$\text{MSPE} = \gamma(0) - 2\mathbf{c}_n^T \boldsymbol{\gamma}_n + \mathbf{c}_n^T \Sigma_n \mathbf{c}_n$$

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Recall that $\mathbf{c}_n = \Sigma_n^{-1} \boldsymbol{\gamma}_n$, therefore we have

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If $\{X_t\}$ is a Gaussian process then an **approximate** $100(1 - \alpha)\%$ **prediction interval** for X_{n+h} is given by

$$P_n X_{n+h} \pm z_{1-\alpha/2} \sqrt{\text{MSPE}}.$$

One-Step Ahead Prediction of AR(1) Process

Consider AR(1) process $X_t = \phi X_{t-1} + Z_t$, where $|\phi| < 1$ and $\{Z_t\} \sim \text{WN}(0, 1 - \phi^2)$.

- ▶ Since $\text{Var}(X_t) = 1$, $\gamma(h) = \rho(h) = \phi^{|h|}$

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- ▶ To forecast X_{n+1} based upon $\mathbf{X}_n = (X_1, \dots, X_n)^T$, using best linear predictor $P_n X_{n+1} = \mathbf{c}_n^T \mathbf{X}_n$, we need to solve $\Sigma_n \mathbf{c}_n = \boldsymbol{\gamma}_n$

$$\begin{bmatrix} 1 & \phi & \dots & \phi^{n-1} \\ \phi & 1 & \dots & \phi^{n-2} \\ \vdots & \vdots & \dots & \vdots \\ \phi^{n-1} & \phi^{n-2} & \dots & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \phi \\ \phi^2 \\ \vdots \\ \phi^n \end{bmatrix}$$

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$$\begin{bmatrix} 1 & \phi & \dots & \phi^{n-1} \\ \phi & 1 & \dots & \phi^{n-2} \\ \vdots & \vdots & \dots & \vdots \\ \phi^{n-1} & \phi^{n-2} & \dots & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \phi \\ \phi^2 \\ \vdots \\ \phi^n \end{bmatrix}$$

$\Rightarrow$ the solution is $\mathbf{c}_n = (\phi, 0, \dots, 0)^T$, yielding

$$P_n X_{n+1} = \mathbf{c}_n^T \mathbf{X}_n = \phi X_n$$

One-Step Ahead Prediction of AR(1) Process (Cont'd)

- ▶ ϕX_n makes intuitive sense as a predictor since

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$$\text{Cov}(Z_{n+1}, X_{n-j+1}) = 0, j = 1, \dots, n$$

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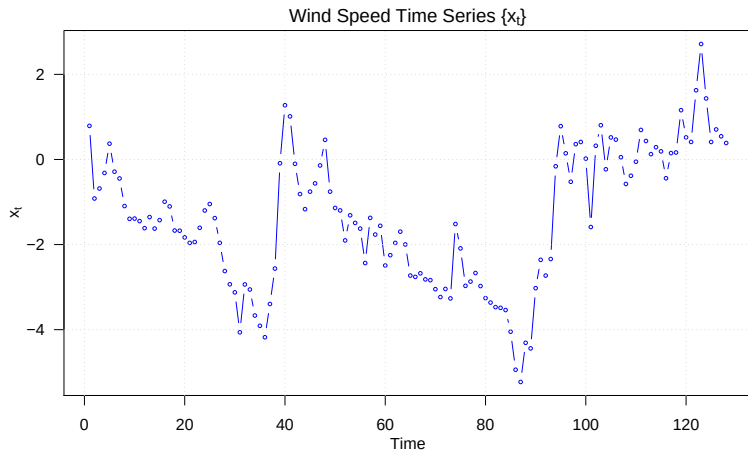
$$\text{Cov}(Z_{n+1}, X_{n-j+1}) = 0, j = 1, \dots, n$$

- ▶ MSPE is

$$\text{Var}(X_{n+1} - \phi X_n) = \gamma(0) - \mathbf{c}_n^T \boldsymbol{\gamma}_n = 1 - \phi^2,$$

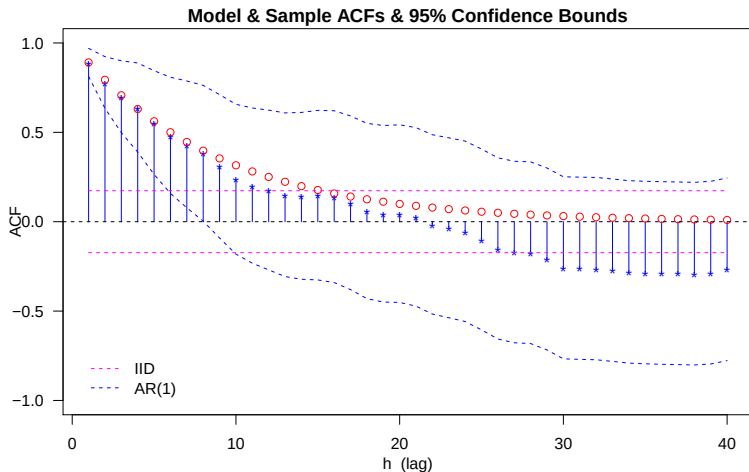
because $\mathbf{c}_n = (\phi, 0, \dots, 0)^T$ and $\boldsymbol{\gamma}_n = (\phi, \phi^2, \dots, \phi^n)^T$

Wind Speed Time Series Example [Source: UW stat 519 lecture notes by Donald Percival]



Let's use this series to illustrate forecasting one step ahead

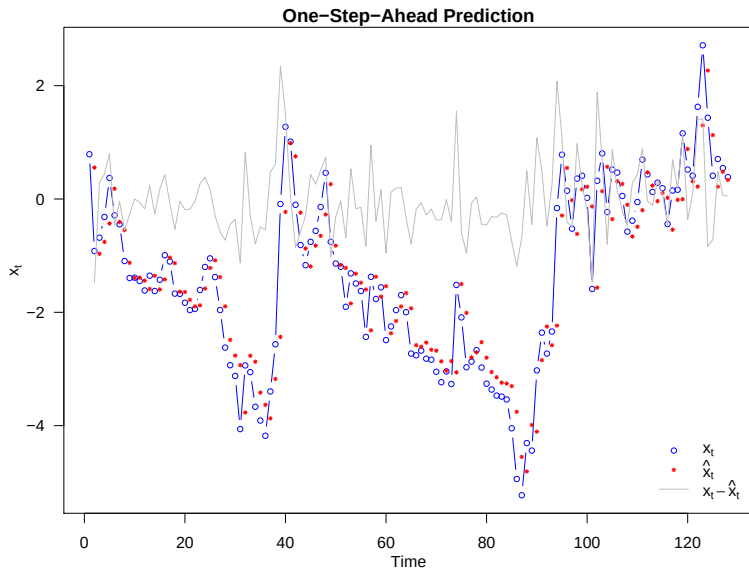
Model & Sample ACFs & 95% Confidence Bounds



The sample ACF indicates compatibility with AR(1) model

$$\Rightarrow P_n X_{n+1} = \phi X_n$$

One-Step-Ahead Prediction of Wind Speed Series



Predicting “Missing” Values

- ▶ Let $\{X_t\}$ be a stationary process with mean μ and ACVF $\gamma(\cdot)$. Suppose we know X_1 and X_3 , and want to predict X_2 using linear combinations of X_1 and X_3

Predicting “Missing” Values

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- ▶ **Solution:** To calculate $P_{X_1, X_3} X_2$ we minimize

$$\begin{aligned}\text{MSPE} &= \mathbb{E} \left[(X_2 - P_{X_1, X_3} X_2)^2 \right] \\ &= \mathbb{E} \left[(X_2 - c_0 - c_1 X_3 - c_2 X_1)^2 \right]\end{aligned}$$

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- ▶ Proceed as for the forecasting case to get the optimal coefficients:
 - ▶ Calculate derivatives

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- ▶ Proceed as for the forecasting case to get the optimal coefficients:
 - ▶ Calculate derivatives
 - ▶ Set the derivatives equal to zero

Predicting “Missing” Values

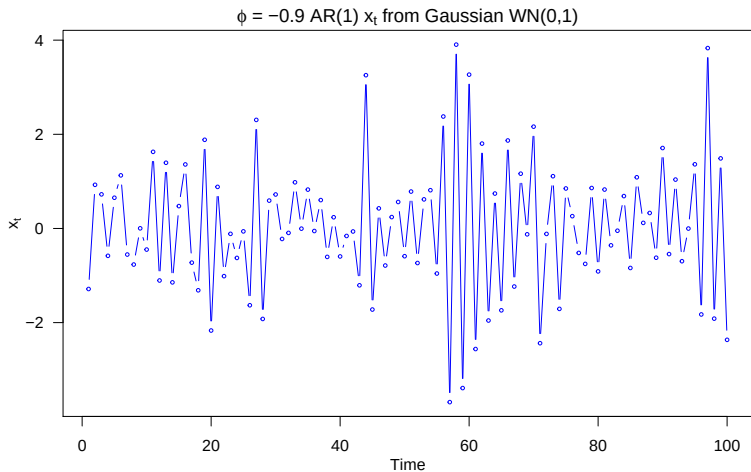
- ▶ Let $\{X_t\}$ be a stationary process with mean μ and ACVF $\gamma(\cdot)$. Suppose we know X_1 and X_3 , and want to predict X_2 using linear combinations of X_1 and X_3

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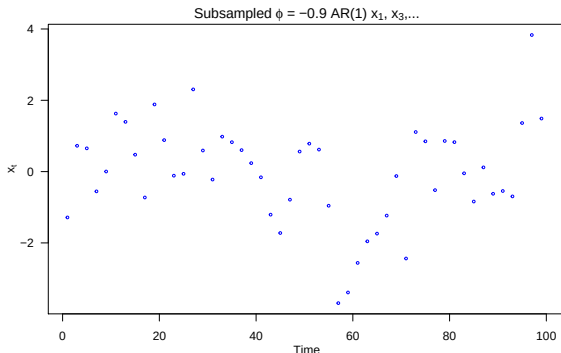
$$\begin{aligned}\text{MSPE} &= \mathbb{E} \left[(X_2 - P_{X_1, X_3} X_2)^2 \right] \\ &= \mathbb{E} \left[(X_2 - c_0 - c_1 X_3 - c_2 X_1)^2 \right]\end{aligned}$$

- ▶ Proceed as for the forecasting case to get the optimal coefficients:
 - ▶ Calculate derivatives
 - ▶ Set the derivatives equal to zero
 - ▶ Solve the linear system of equations

Another AR(1) Example with $\phi = -0.9$



Subsampled $X_1, X_3, \dots$ and Removed $X_2, X_4, \dots$



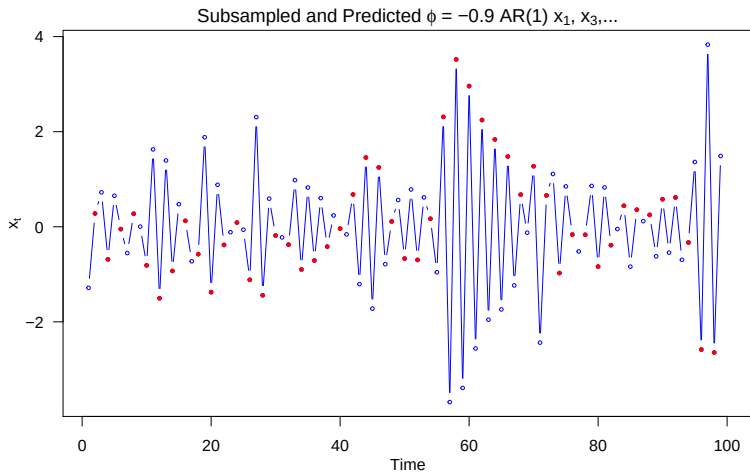
The best linear predictor of X_2 given X_1, X_3 is

$$\hat{X}_2 = \frac{\phi}{1 + \phi^2} (X_1 + X_3),$$

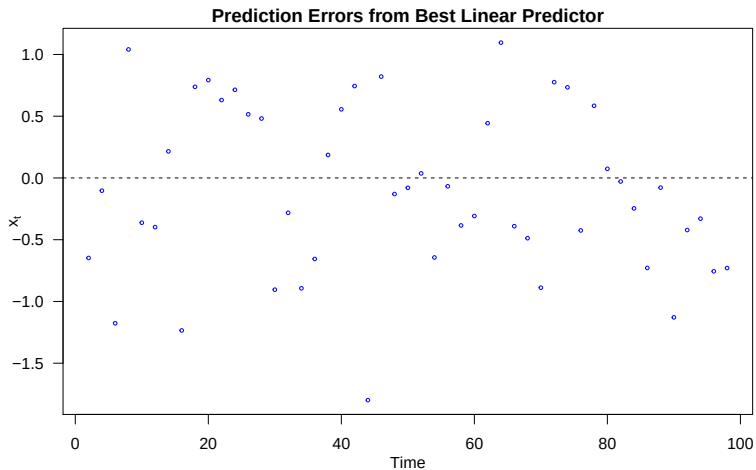
and the MSPE is

$$\frac{\sigma^2}{1 + \phi^2}$$

Predict $X_2, X_4, \dots$ Using Best Linear Predictor



Prediction Errors from Best Linear Predictor



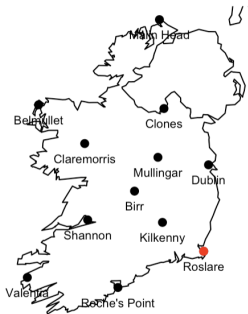
A Modeling Case Study of Ireland Wind Data

(Courtesy of Peter Craigmile's time series lecture notes)

Data Description [Haslett & Raftery, 1989 ¹]

Twelve wind stations collected daily readings over 18 years (from 1961 to 1978). Wind speeds were measured in knots (1 knot = $0.5148 \frac{m}{s}$)

We will focus on the wind data from 1965-1969 at the Rosslare station

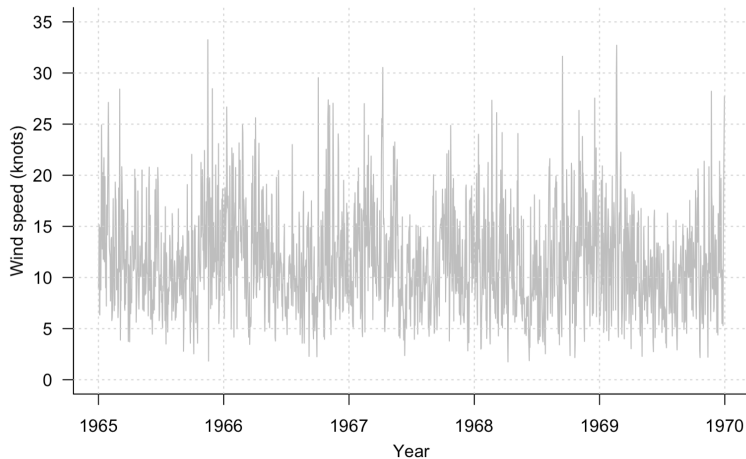


Modeling procedure:

- ▶ Exploratory analysis
- ▶ Model and remove the trend and seasonal components
- ▶ ARMA model identification, fitting, and selection
- ▶ Perform forecast

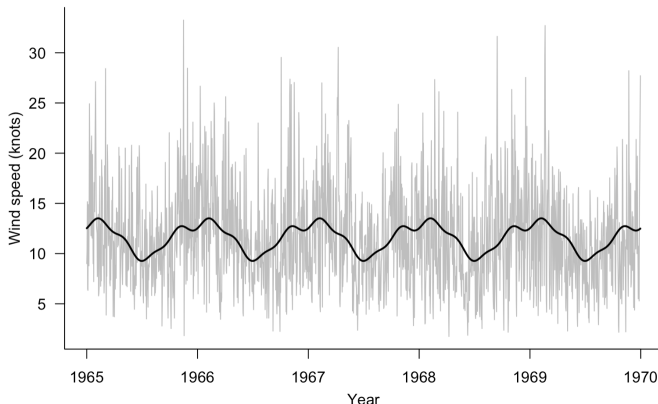
¹Haslett, J., & Raftery, A. E. (1989). Space-time modelling with long-memory dependence: Assessing Ireland's wind power resource. *Journal of the Royal Statistical Society: Series C*, 38(1), 1-21.

Wind Speed Time Series at Rosslare Station



- ▶ No clear trend
- ▶ Seasonal Pattern

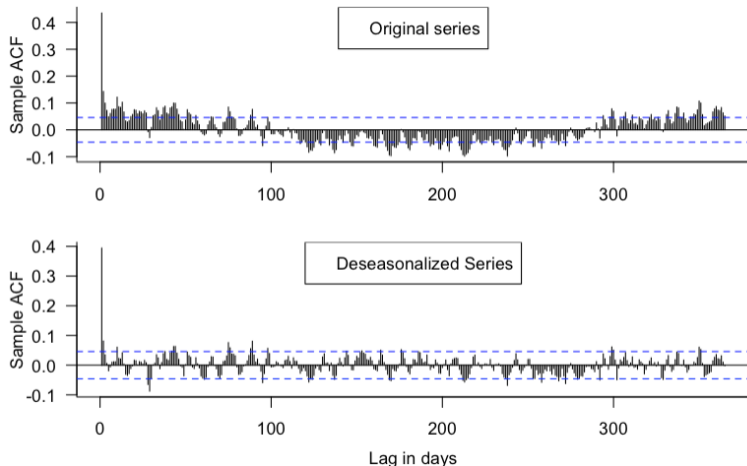
Estimating the Seasonal Pattern



Here we use **harmonic regression** with 4 harmonics per year to model the seasonal components

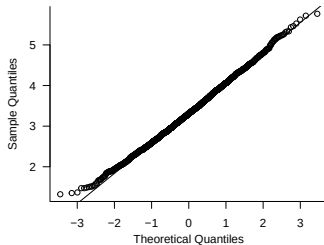
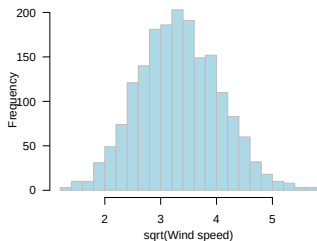
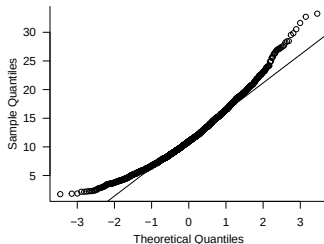
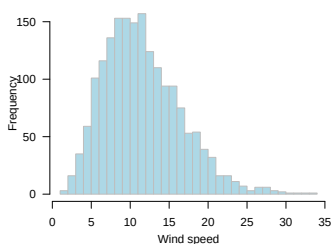
$$s_t = \beta_0 + \sum_{j=1}^4 (\beta_{1j} \cos(2\pi jt) + \beta_{2j} \sin(2\pi jt))$$

ACF Plots: Original and Deseasonalized Series



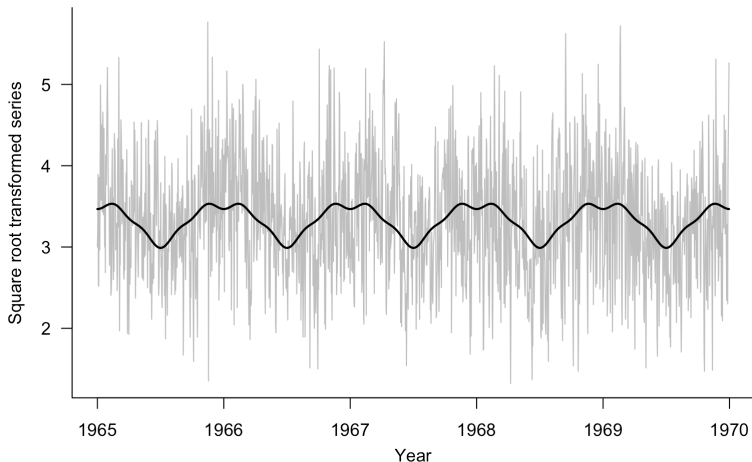
Seasonal modeling (via harmonic regression) effectively removes the oscillatory pattern in the ACF of the original series

Transform Data to Approximate Gaussian Distribution



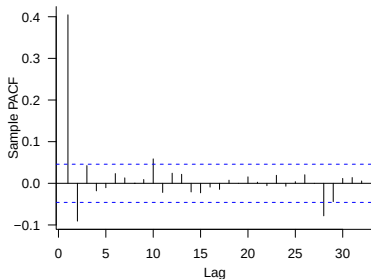
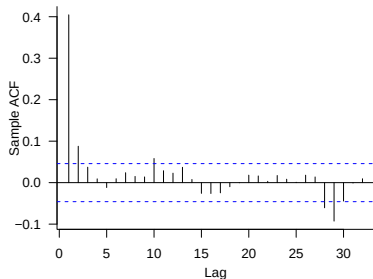
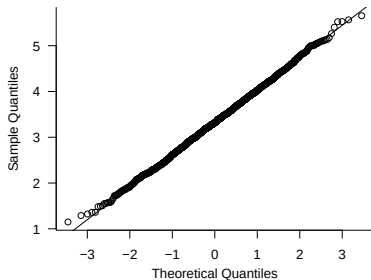
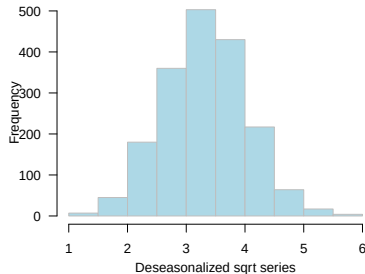
The square-root transformation improves the Gaussian approximation. Now take the square root of the original data and deseasonalize again!

Estimating Transformed Series Seasonality



Next, we need to check whether the deseasonalized series is approximately Gaussian

Marginal and ACF/PACF of the Deseasonalized Series



Based on ACF/PACF, which ARMA model would you choose?

Potential Model 1: AR(1)

```
> ar1.model <- arima(sqrt.rosslare.ds, order = c(1, 0, 0))  
> ar1.model
```

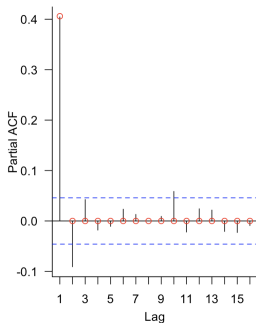
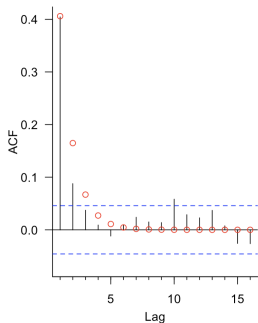
Call:

```
arima(x = sqrt.rosslare.ds, order = c(1, 0, 0))
```

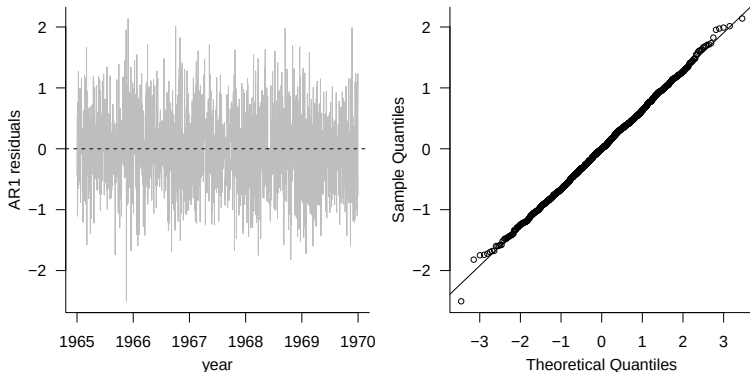
Coefficients:

	ar1	intercept
	0.4060	3.3257
s.e.	0.0214	0.0254

σ^2 estimated as 0.4148: log likelihood = -1787.72, aic = 3581.43



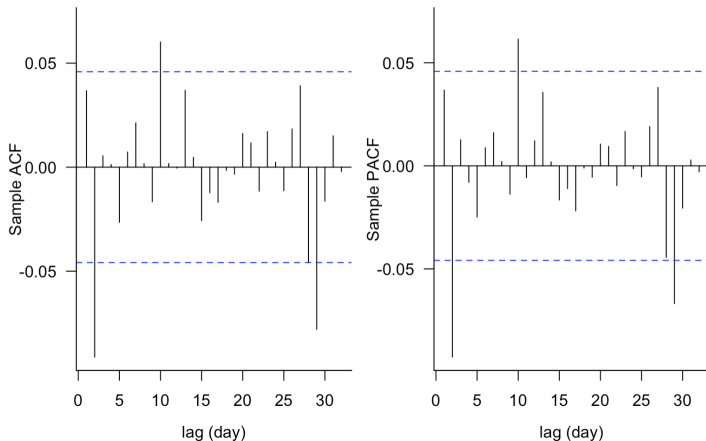
Residual Plots for the AR(1) Model



Normality assumption seems reasonable.

Next check the [ACF/PACF](#) and perform a [Box test](#) to assess if the AR(1) fit adequately accounts for the temporal dependence structure

Diagnostic for the AR(1) Model



```
> Box.test(ar1.resids, lag = 32, fitdf = 1, type = "Ljung-Box")
```

Box-Ljung test

```
data: ar1.resids
```

```
X-squared = 53.142, df = 31, p-value = 0.00794
```

Potential Model 2: AR(2)

```
> (ar2.model <- arima(sqrt.rosslare.ds, order = c(2, 0, 0)))
```

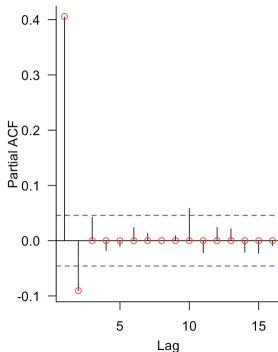
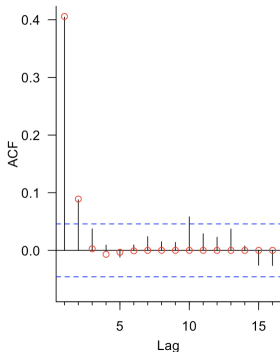
Call:

```
arima(x = sqrt.rosslare.ds, order = c(2, 0, 0))
```

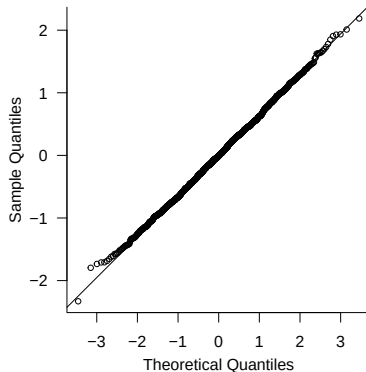
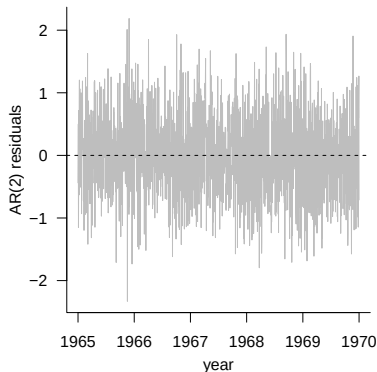
Coefficients:

	ar1	ar2	intercept
	0.4425	-0.0905	3.3254
s.e.	0.0233	0.0233	0.0232

σ^2 estimated as 0.4114: log likelihood = -1780.23, aic = 3568.46



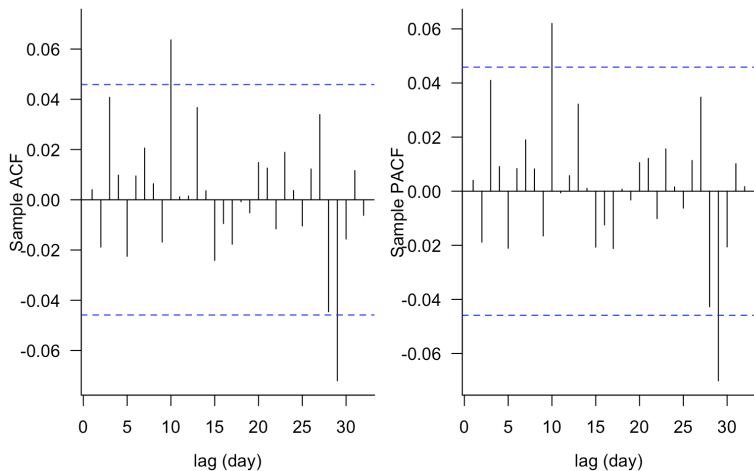
Residual Plots for the AR(2) Model



Normality assumption seems reasonable.

Next check the [ACF/PACF](#) and perform a [Box test](#) to assess if the AR(2) fit adequately accounts for the temporal dependence structure

Diagnostic for the AR(2) Model



```
> Box.test(ar2.resids, lag = 32, fitdf = 2, type = "Ljung-Box")
```

Box-Ljung test

```
data: ar2.resids
```

```
X-squared = 36.548, df = 30, p-value = 0.1907
```

Potential Model 3: ARMA(1, 1)

```
> (arma11.model <- arima(sqrt.rosllare.ds, order = c(1, 0, 1)))
```

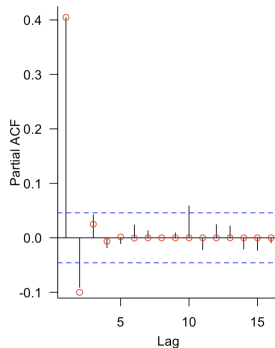
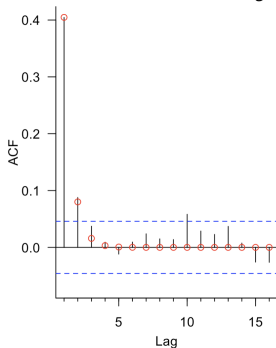
Call:

```
arima(x = sqrt.rosllare.ds, order = c(1, 0, 1))
```

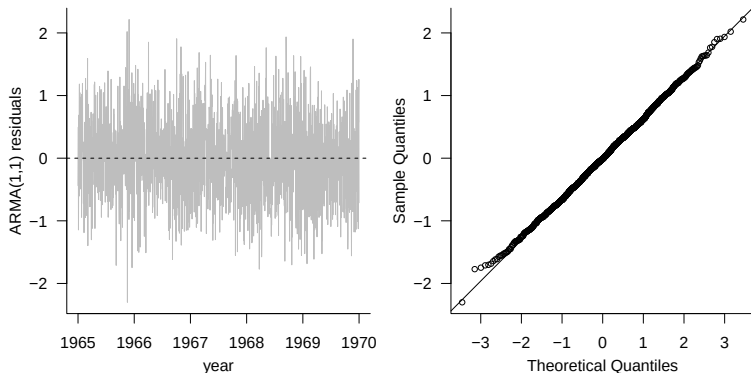
Coefficients:

	ar1	ma1	intercept
	0.1978	0.2502	3.3254
s.e.	0.0556	0.0553	0.0234

sigma² estimated as 0.4108: log likelihood = -1778.82, aic = 3565.64



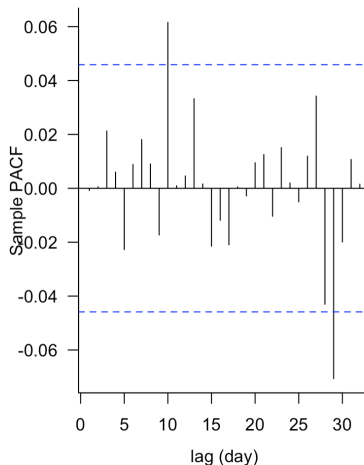
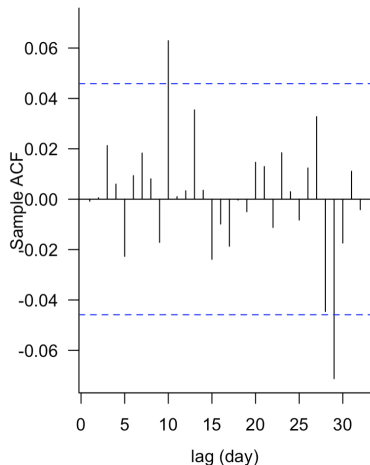
Residual Plots for the ARMA(1, 1) Model



Normality assumption seems reasonable.

Next check the [ACF/PACF](#) and perform a [Box test](#) to assess if the ARMA(1, 1) fit adequately accounts for the temporal dependence structure

Diagnostic for the ARMA(1, 1) Model



```
> Box.test(arma11.resids, lag = 32, fitdf = 2, type = "Ljung-Box")
```

Box-Ljung test

data: arma11.resids

X-squared = 32.757, df = 30, p-value = 0.3332

Potential Model 4: ARMA(2, 1)

```
> (arma21.model <- arima(sqrt.rosslare.ds, order = c(2, 0, 1)))
```

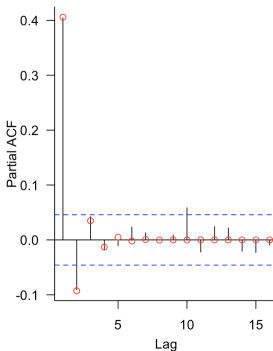
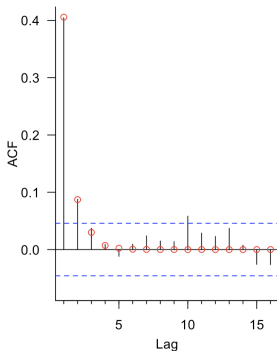
Call:

```
arima(x = sqrt.rosslare.ds, order = c(2, 0, 1))
```

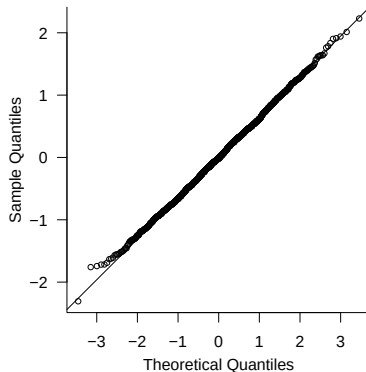
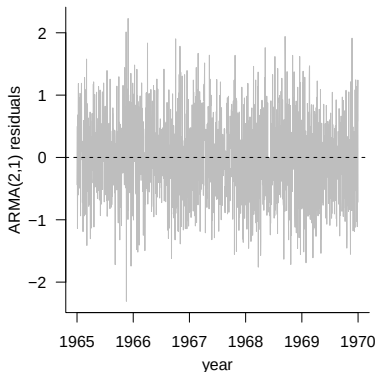
Coefficients:

	ar1	ar2	ma1	intercept
	0.0703	0.0587	0.3768	3.3253
s.e.	0.1691	0.0772	0.1663	0.0237

σ^2 estimated as 0.4107: log likelihood = -1778.56, aic = 3567.11



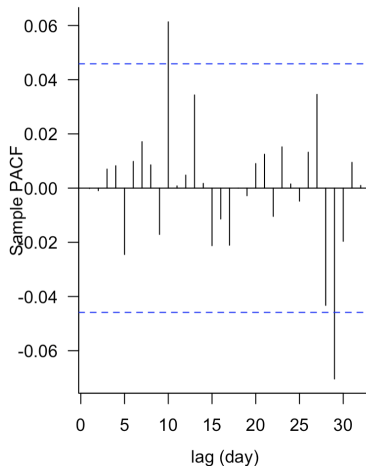
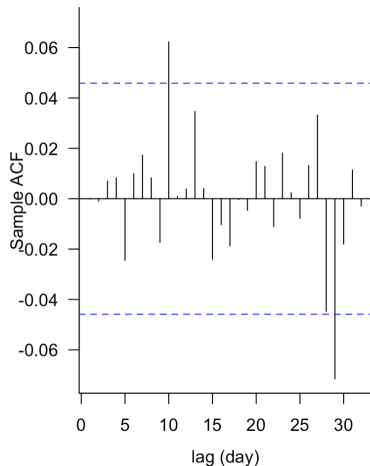
Residual Plots for the ARMA(2, 1) Model



Normality assumption seems reasonable.

Next check the [ACF/PACF](#) and perform a [Box test](#) to assess if the ARMA(2, 1) fit adequately accounts for the temporal dependence structure

Diagnostic for the ARMA(2, 1) Model



```
> Box.test(arma21.resids, lag = 32, fitdf = 3, type = "Ljung-Box")
```

Box-Ljung test

```
data: arma21.resids
```

```
X-squared = 32.171, df = 29, p-value = 0.3124
```

Comparing Models via Information Criteria

Model	AIC	AICc
AR(1)	3583.817	3583.824
AR(2)	3570.650	3570.663
ARMA(1, 1)	3567.833	3567.847
ARMA(2, 1)	3569.319	3569.341

Which model would you pick?

Forecasting Future Wind Speeds

Question: How do we predict wind speeds on the original scale, including the seasonality that was previously estimated?

- ▶ Suppose we want to predict the next 7 days of wind speed values. We base our forecasts on the chosen ARMA(1,1) model.
- ▶ We need to reverse the order of our modeling process: $\Rightarrow$ forecast under the transformed scale $\rightarrow$ add the estimated seasonal component $\rightarrow$ back-transform to the original scale.

Forecasting Future Wind Speeds, continued

- The **forecasts** for the next 7 days of deseasonalized square root values are:

```
> round(sqrt.rosslare.forecast$pred, 3)
```

Time Series:

Start = c(1970, 1)

End = c(1970, 7)

Frequency = 365

```
[1] 3.997 3.458 3.352 3.331 3.326 3.326 3.325
```

- The **standard error** for the forecasts are:

```
> round(sqrt.rosslare.forecast$se, 3)
```

Time Series:

Start = c(1970, 1)

End = c(1970, 7)

Frequency = 365

```
[1] 0.641 0.702 0.705 0.705 0.705 0.705 0.705
```

Forecasting Future Wind Speeds, continued

- ▶ Next, we add back in the seasonality to get:

```
> adj.forecast <- fitted(harm.model)[1:h] + sqrt.rosslare.forecast$pred  
> round(adj.forecast, 3)
```

Time Series:

Start = c(1970, 1)

End = c(1970, 7)

Frequency = 365

1	2	3	4	5	6	7
4.139	3.600	3.494	3.473	3.470	3.470	3.470

- ▶ Finally, we transform back to the original scale

```
> round(adj.forecast^2, 3)
```

Time Series:

Start = c(1970, 1)

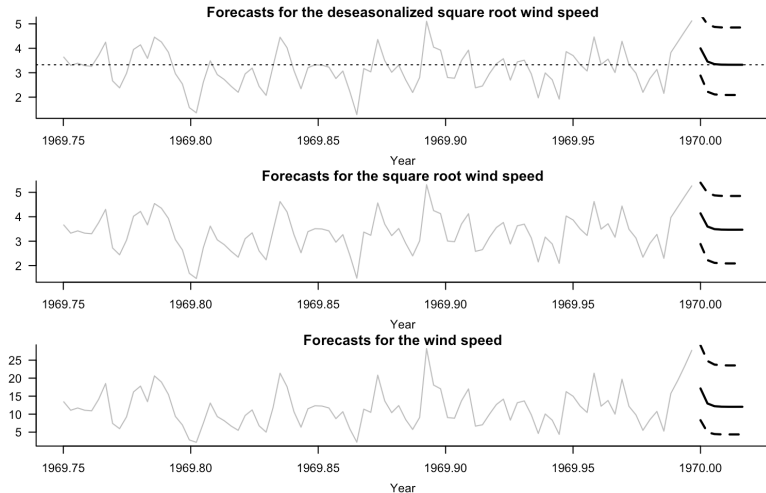
End = c(1970, 7)

Frequency = 365

1	2	3	4	5	6	7
17.132	12.962	12.208	12.064	12.039	12.039	12.044

- ▶ To get the prediction limits, we need to transform the lower and upper prediction limits on the sqrt scale

Visualizing the Forecasts



Further Questions

- ▶ What is the full model for our time series data?
- ▶ Is there a better description for the trend than just a constant term? What about alternative seasonal modeling?
- ▶ How well do we forecast? What about forecast uncertainty?